Resource-Constrained Cognitive Path Optimization: From Local Plasticity to Low-Cost Predictive Inference

Qing Zhou

Abstract

This paper proposes a resource-constrained cognitive path optimization framework. The brain is not treated as an unlimited computational system searching for absolute truth, but as a finite-resource adaptive system settling into cognitive paths that reduce future uncertainty, support action, and preserve adaptive self-environment coupling under constraints of energy, attention, time, and memory capacity. The formal object is not a physical action functional in the Hamiltonian sense. Rather, the paper develops a resource-constrained active inference framework whose central object is a path-cost functional for dissipative cognitive dynamics. In this framework, predictive coding supplies the error-cancellation mechanism, neural integration supplies a microscopic dynamical basis, synaptic plasticity and adaptive thresholds turn repeated stimulation into priors, variational free energy supplies an observer-level cost function, resource constraints generate a path-level cost functional across time, multi-model Bayesian competition explains the dynamic selection among candidate interpretations, memory dynamics preserve reusable low-cost paths through local plasticity, and the higher-order observer is formalized as a meta-control system that modulates cost weights, precision, update gains, and effective temperature.

Keywords: active inference; free energy principle; cognitive path optimization; path-cost functional; memory dynamics; resource constraints; predictive processing; local plasticity; meta-control; abstraction; cognitive overload

1 Starting Point

In classical physics, the principle of least action states that the actual trajectory of a system is not arbitrary. Under suitable constraints, the system follows a path for which an action functional is stationary or optimal. The present paper borrows only the broad intuition that trajectories can be compared by accumulated path cost.

An analogous, but not identical, idea can be introduced for cognition:

> The brain is not an unlimited computational system. It is a finite-resource system

that selects cognitive paths that are sufficiently stable, sufficiently low-cost, able to explain the world, and useful for action.

Here the central object is not physical action, the integral of kinetic minus potential energy. It is a cognitive path-cost functional: the accumulated cost paid by a cognitive system across time. This cost includes prediction error, neural energy consumption, attentional load, memory maintenance cost, higher-order control cost, and action cost.

In this sense, the brain may be understood as a resource-constrained path-optimization system. More precisely:

> Intelligence is not unlimited computation, but path selection under finite resources.

2 Relation to Existing Work

The present framework is not intended to replace existing theories of bounded cognition, cognitive control, compression, or active inference. Its purpose is to connect them through a single path-cost formulation.

First, the finite-resource assumption is closely related to resource-rational analysis, especially the work of Lieder and Griffiths. Resource-rational analysis extends Bayesian and rational models of cognition by asking which cognitive operations are available to an agent and how costly those operations are in time and computational resources. The present model inherits this spirit, but embeds resource costs inside an active-inference path functional rather than treating them only as constraints on isolated computations.

Second, the treatment of higher-order control is closely aligned with the Expected Value of Control theory developed by Shenhav, Botvinick, and Cohen. EVC proposes that control allocation depends on expected payoff, the amount of control required, and the cognitive effort cost. In the present framework, the term $\lambda_C C_t$ and the meta-control variable θ_t generalize this idea into a temporally extended path-selection model.

Third, the claim that abstraction is a compressed reusable structure resonates with minimum description length and compression-based theories of learning and curiosity. Rissanen's MDL principle treats good models as those that shorten the description of data. Schmidhuber's theory of compression progress treats improvements in compressibility as a driver of curiosity and discovery. The present framework adds a memory-dynamical interpretation: abstractions persist because repeated co-activation and consolidation leave reusable circuits whose observer-level effect is to reduce future free energy across multiple contexts by more than their maintenance cost.

Finally, the free energy principle and active inference provide the formal background for interpreting perception, action, attention, and learning as forms of uncertainty reduction. The contribution of this paper is to combine active inference with resource-rational costs, control valuation, memory retention, abstraction, and overload into a single resource-constrained cognitive path optimization model.

3 From Free Energy to Cognitive Path Cost

In the free energy principle and active inference, a system can be modeled as minimizing variational free energy in order to make its internal beliefs approximate the generative structure of the external world. This is a macro-level and normative description. It does not imply that individual neurons explicitly compute a free-energy functional or compare future values.

Let the hidden environmental state be

$$x_t \in \mathcal{X}, \tag{1}$$

and the observation be

$$o_t \in \mathcal{O}. \tag{2}$$

The system maintains an approximate posterior

$$q(x_t). \tag{3}$$

For a generative model m^k, the instantaneous variational free energy can be written as

$$F_t^k[q^k] = \mathbb{E}_{q^k}\left[\ln q^k(x_t) - \ln p(o_t, x_t|o_{1:t-1}, m^k)\right]. \tag{4}$$

It can also be decomposed as

$$F_t^k[q^k] = D_{KL}\left[q^k(x_t)\|p(x_t|o_{1:t}, m^k)\right] - \ln p(o_t|o_{1:t-1}, m^k). \tag{5}$$

Since the KL divergence is non-negative,

$$F_t^k[q^k] \geq -\ln p(o_t|o_{1:t-1}, m^k). \tag{6}$$

Thus, minimizing variational free energy makes the approximate posterior closer to the Bayesian posterior and increases the explanatory adequacy of the model for the current observation. At the biological implementation level, the corresponding mechanism may be local error signaling, repeated activation, synaptic plasticity, threshold adaptation, and neuromodulatory consolidation. Free energy is the modeling language used to describe the aggregate effect of these local processes.

However, the brain does not minimize error only at an isolated instant. It must maintain perception, memory, action, and control continuously over time. The instantaneous cost must therefore be extended into a path-level cost.

Expected free energy is normally a future-oriented quantity. To avoid mixing an instantaneous present-time free energy term with a future policy evaluation term, define a rolling-horizon expected free energy:

$$G_{t:t+H}(\pi_t) = \int_t^{t+H} G_\tau(\pi_t)d\tau. \tag{7}$$

In the path functional below, let

$$\mathcal{G}_t(\pi_t) = \frac{1}{H}G_{t:t+H}(\pi_t) \tag{8}$$

denote the local expected free-energy rate associated with the policy considered at time t. This keeps the integrand dimensionally and temporally consistent.

Define a resource-constrained cognitive path-cost functional:

$$\mathcal{C}_{path}[q,\pi] = \int_{t_0}^{t_1} \left[F_t(q_t) + \beta\mathcal{G}_t(\pi_t) + \lambda_E E_t + \lambda_A A_t + \lambda_M M_t + \lambda_C C_t\right] dt. \tag{9}$$

Here F_t is current variational free energy; $\mathcal{G}_t(\pi_t)$ is the local expected free-energy rate induced by a rolling future horizon; E_t is energetic cost; A_t is attentional cost; M_t is memory maintenance and retrieval cost; C_t is higher-order control cost; and $\lambda_E, \lambda_A, \lambda_M, \lambda_C$ are resource cost weights.

The energetic term should not be read as an independent penalty merely appended to free energy. In a predictive-processing interpretation, energetic cost is partly generated by the neural activity required to encode prediction error. If a model m^k produces a prediction $\hat{o}_t^k$ and a prediction error

$$\varepsilon_t^k = o_t - \hat{o}_t^k, \tag{10}$$

then a coarse operational form is

$$E_t^k = E_{\text{base}} + \chi(\varepsilon_t^k)^\top \Pi_t^k \varepsilon_t^k + E_{\text{plastic}}^k(t), \tag{11}$$

where Π_t^k is precision and E_{plastic}^k is the short-term cost of changing synapses or thresholds. Equivalently, at the information-theoretic level, the activity cost may be treated as increasing with surprisal:

$$E_{\text{act}}^k(t) \propto -\ln p(o_t|m^k, o_{1:t-1}). \tag{12}$$

Thus, as repeated stimulation leaves a stronger circuit trace, the modeler can describe the same process as $p(o_t|m^k, o_{1:t-1})$ rising, prediction error falling, and activation cost approaching a low baseline. This coupling prevents the model from treating free-energy reduction and energy saving as unrelated bookkeeping terms, while preserving the distinction between local neural adaptation and observer-level Bayesian description.

Cognitive path selection can then be written as

$$(q^*, \pi^*) = \arg\min_{q,\pi} \mathcal{C}_{path}[q, \pi]. \tag{13}$$

This is not a direct equivalence to physical action. It is a path-cost functional for a resource-constrained active inference system.

4 The Brain Maintains Multiple Candidate Paths

Cognition does not usually begin with a single final answer. In complex situations, the brain maintains several candidate interpretations:

$$H_1, H_2, \ldots, H_K. \tag{14}$$

Each candidate interpretation corresponds to a generative model m^k and has a belief weight

$$\rho_t^k = p(m^k | o_{1:t}), \tag{15}$$

with

$$\rho_t^k \geq 0, \qquad \sum_{k=1}^{K} \rho_t^k = 1. \tag{16}$$

The overall belief can be represented as a mixture:

$$q(x_t) = \sum_{k=1}^{K} \rho_t^k q^k(x_t). \tag{17}$$

If each path has a total instantaneous cost

$$\ell_t^k = F_t^k + \beta \mathcal{G}_t^k + \lambda_E E_t^k + \lambda_A A_t^k + \lambda_M M_t^k + \lambda_C C_t^k, \tag{18}$$

then model weights can be updated by an exponential rule:

$$\rho_{t+\Delta t}^k = \frac{\rho_t^k \exp(-\ell_t^k \Delta t)}{\sum_j \rho_t^j \exp(-\ell_t^j \Delta t)}. \tag{19}$$

Taking the continuous-time limit $\Delta t \to 0$ yields the replicator equation:

$$\frac{d\rho_t^k}{dt} = \rho_t^k (\bar{\ell}_t - \ell_t^k), \tag{20}$$

where

$$\bar{\ell}_t = \sum_j \rho_t^j \ell_t^j. \tag{21}$$

This means:

> Interpretive paths whose cost is below the average gain weight, while paths whose cost is above the average lose weight.

This helps explain several cognitive phenomena: hesitation, insight, association, narrowed thinking under fatigue, and reduced abstraction under stress. The cognitive state of the brain is therefore not a single fixed answer. It is a dynamic competition among candidate interpretations under finite resource constraints.

5 Neural Integration as a Microscopic Basis

A single neuron can be described, in a simplified form, as a temporal integrator:

$$\tau \frac{dV}{dt} = -(V - V_{rest}) + I(t). \tag{22}$$

When membrane potential reaches threshold,

$$V(t) \geq V_{th}, \tag{23}$$

the neuron fires.

Thus the neuron is already a time-integration and threshold-triggering system:

$$\text{input accumulation} \rightarrow \text{threshold crossing} \rightarrow \text{spiking}. \tag{24}$$

This equation should not be overread. LIF dynamics can anchor the microscopic description of memory dynamics, but they do not by themselves derive memory, priors, value, dopamine, free energy, abstraction, or meta-control. Those require additional modeling assumptions: an error-coding assumption, a plasticity rule, a homeostatic regulation rule, a value-gating rule, and a coarse-graining map from neural populations to mnemonic variables.

5.1 From LIF to Memory Dynamics: Conditional Derivation

The conditional bridge begins by separating raw drive from prediction-error drive. In a predictive-coding circuit, the effective drive is not merely the bottom-up input; it is the mismatch between input and a generative prediction. Let a lower-level signal be y_t and let a higher-level memory or prior generate the prediction

$$\hat{y}_t = G(m_t, W_t). \tag{25}$$

The error population carries

$$\varepsilon_t = y_t - \hat{y}_t, \tag{26}$$

but the system should not be read as if every neuron directly receives the full vector ε_t. A more explicit two-population approximation distinguishes error units from state or belief units:

$$\tau_\varepsilon \dot{V}_\varepsilon = -(V_\varepsilon - V_{\text{rest}}) + A\left(y_t - G(m_t, W_t)\right) + \zeta_\varepsilon, \tag{27}$$

and

$$\tau_m \dot{V}_m = -(V_m - V_{\text{rest}}) + B\Pi_t \varepsilon_t + I_{\text{value}}(t) + \zeta_m. \tag{28}$$

Here Π_t is precision. What drives belief updating is not only bare error but precision-weighted prediction error. On first exposure, m_t and W_t do not yet predict the signal well, so ε_t is large and error-unit activity, control demand, and plasticity pressure are high. After repeated exposure, the predictable part of the input is explained by the top-down model. This does not imply that all neural activity disappears; rather, prediction-error traffic and effortful control usually decline while representational, motor, or habit circuits may remain efficiently active.

Learning also changes the future initial conditions for activation. A minimal plasticity equation is

$$\frac{dW_{ij}}{dt} = \eta_W \,\text{pre}_j(t)\,\text{post}_i(t) - \lambda_W W_{ij}, \tag{29}$$

which should be read as a rate-based Hebbian approximation. In spike-based models this term can be replaced by an STDP rule whose sign and magnitude depend on the timing difference between presynaptic and postsynaptic spikes. In an error-correcting predictive-coding form, if $\hat{y} = W\phi(m)$ and

$$F_t = \frac{1}{2}\varepsilon_t^\top \Pi_t \varepsilon_t + \frac{1}{2}\lambda_W \|W\|^2, \tag{30}$$

then gradient descent on F_t gives the local update

$$\dot{W} = \eta_W \,\Pi_t \varepsilon_t\, \phi(m_t)^\top - \eta_W \lambda_W W. \tag{31}$$

Not every error should be learned. A value- and salience-sensitive plasticity gate can be written as

$$q_t = \sigma\left(a_\varepsilon \|\Pi_t \varepsilon_t\| + a_V V_{\text{value},t} + a_N \text{Novelty}_t - a_E E_{\text{plastic},t}\right), \tag{32}$$

so that

$$\dot{W} = \eta_W q_t \,\Pi_t \varepsilon_t\, \phi(m_t)^\top - \lambda_W W. \tag{33}$$

This gate is a compact model of neuromodulatory control over learning rate and consolidation; it should not be reduced to dopamine alone.

The threshold is also adaptive. A homeostatic intrinsic-plasticity rule can be written as

$$\tau_{\text{th}} \frac{dV_{\text{th}}}{dt} = \kappa_{\text{th}} \left(r_t - r_{\text{target}}\right), \tag{34}$$

where r_t is recent firing rate. Sustained over-firing raises the threshold, while under-firing lowers it. This rule is not the main mechanism that filters predicted input; prediction-error coding performs that filtering. Homeostatic threshold adaptation keeps the circuit from becoming either explosively excitable or silent. Hebbian plasticity makes useful pathways easier to recruit; homeostatic plasticity keeps that reinforcement bounded.

At the cognitive level, the activation strength b of a belief, memory, or circuit can be described by a coarse-grained phenomenological model:

$$\tau\dot{b} = -\alpha b - \Gamma_b \frac{\partial F}{\partial b} + \gamma V_{value} - \eta E + \xi_t. \tag{35}$$

Here b is the activation strength of a belief or circuit; $-\partial F/\partial b$ is the free-energy gradient acting on that activation; V_{value} is a value signal; E is energetic or resource cost; $\Gamma_b, \alpha, \gamma, \eta$ are regulatory parameters; and ξ_t is unmodeled noise or fluctuation.

Because prediction errors are often the main components of free-energy gradients in predictive processing models, this can be approximated heuristically as

$$\tau_b\dot{b} \approx -\alpha b + \beta\Pi_t\varepsilon_t + \gamma V_{\text{value}} - \eta E_{\text{act}} + \xi_t. \tag{36}$$

This second expression should be read as a phenomenological summary, not as a strict derivation from the free energy principle. A more closed-loop summary is

$$\tau_b\dot{b} \approx -\alpha b + \beta\Pi_t\varepsilon(m, W) + \gamma V_{\text{value}} - \eta E_{\text{act}}(\varepsilon) + \xi_t, \tag{37}$$

with

$$E_{\text{act}}(\varepsilon) = E_{\text{base}} + \chi\, \varepsilon^{\top}\Pi\varepsilon. \tag{38}$$

Because the same quadratic prediction-error term may already appear inside F_t, E_{act} should be interpreted as metabolic, control, or plasticity cost rather than a second copy of the same inferential objective.

The memory variable itself is therefore a coarse-grained population-level object:

$$m_i(t) = \mathcal{C}_i\left(\{V_j(t), s_j(t), W_{jk}(t), V_{\text{th},j}(t)\}_{j,k\in\Omega_i}\right). \tag{39}$$

Under this coarse-graining, repeated experience produces the closed loop

$$m_t, W_t \to \hat{y}_t \to \varepsilon_t \to \Delta W_t, \Delta m_t \to m_{t+1}, W_{t+1}. \tag{40}$$

This suggests:

> The nervous system integrates prediction errors, value, plasticity, thresholds, and energetic costs over time in order to determine which circuits are activated, reinforced, inhibited, or forgotten.

Neural firing integration can therefore be understood as a microscopic dynamical basis for free-energy minimization and cognitive path selection only in this conditional sense. LIF supplies the integration-threshold skeleton; predictive coding defines what counts as effective error; plasticity changes future predictions; homeostatic regulation stabilizes the circuit; value gates decide which errors are worth consolidating; and coarse-graining turns population activity into memory variables. Memory and prior formation are not external additions to this dynamics. They are the closed-loop process by which repeated error changes W_t, m_t, excitability, and thresholds, reducing future prediction error and the cost of processing similar inputs.

6 Memory Is Not Recording, but the Preservation of Low-Cost Paths

The brain does not compute everything from scratch. It preserves useful paths from the past so that similar situations can be handled at lower cost in the future.

Memory is therefore not a complete recording of the past. It is a compressed, selective, and reconstructive preservation of past cognitive paths.

Let the strength of a memory trace be

$$m_i(t) \in [0, 1]. \tag{41}$$

At the neural implementation level, its dynamics should be written in local variables, not in terms of an explicitly computed future free-energy benefit. A minimal local rule is

$$dm_i = [-\lambda_i m_i + \alpha_a a_i(t) + \alpha_c \, \mathrm{coact}_i(t) + \alpha_n n_i(t)] \, dt + \sigma_m dW_t^i, \tag{42}$$

where $a_i(t)$ is recent activation of the trace; $\mathrm{coact}_i(t)$ is local pre-post or circuit-level co-activation; $n_i(t)$ is a neuromodulatory or consolidation signal, such as salience, reward, arousal, sleep-dependent replay, or biological availability for plasticity; and $\lambda_i m_i$ is decay or competition from unused traces.

The same process can be described at the observer level by defining

$$\Delta F_i^{obs}(t) = F_{\mathrm{baseline}}(t) - F_{\mathrm{with}\ m_i}(t). \tag{43}$$

This is not a quantity computed by the neuron. It is the reduction in free energy that an external modeler attributes to the presence of the trace. If $\Delta F_i^{obs}(t) > 0$, the trace helps the model explain the current situation; if $\Delta F_i^{obs}(t) \leq 0$, it provides no observer-level explanatory benefit and may even interfere.

For the trace to function as a prior, it must also alter the circuit that will process similar inputs later. One can express this coupling as

$$\text{repeated co-activation} \uparrow \implies W_i \uparrow,\ V_{\text{th},i} \text{ adapts} \implies \varepsilon_t(o_t, m_i) \downarrow,\ E_{\text{act}}(o_t|m_i) \downarrow. \quad (44)$$

Equivalently, at the observer level,

$$E_{\text{act}}(o_t|m_i) = E_{\text{base}} + \chi[-\ln p(o_t|m_i)], \quad (45)$$

or, for Gaussian predictive coding,

$$E_{\text{act}}(o_t|m_i) = E_{\text{base}} + \chi\, \varepsilon_t^\top \Pi_t \varepsilon_t. \quad (46)$$

The memory trace therefore has two linked effects. Locally, repeated stimulation changes connection strength, excitability, and thresholds. Macroscopically, the resulting circuit looks as if it lowers future free energy by improving prediction, and it physically lowers future metabolic cost by reducing the error signals that need to be fired.

The longer-term retention rule should therefore be read as an observer-level summary of local plasticity rather than a neural decision rule. Define an effective mnemonic reward as

$$r_i^{mem,obs}(t) = \max(0, \Delta F_i^{obs}(t)) - c_i(m_i, t), \quad (47)$$

where $c_i(m_i, t)$ is the ongoing maintenance cost of the trace. Then the observer-level value of retaining the memory can be written as

$$V_i^{obs}(t) = \mathbb{E}\left[\int_t^\infty e^{-\gamma(s-t)} r_i^{mem,obs}(s) ds\right]. \quad (48)$$

The retention rule for a memory can be written as a macro-level equivalence:

$$\text{trace tends to remain reusable} \iff V_i^{obs}(t) > 0, \quad (49)$$

or equivalently,

$$\mathbb{E}\left[\int_t^\infty e^{-\gamma(s-t)} \Delta F_i^{obs}(s) ds\right] > C_{\text{maintain}}(m_i). \quad (50)$$

Thus:

> What remains reusable in the long run is not chosen by a neuron because it predicts future benefit. It is the structure repeatedly reactivated and consolidated by local physiology; from the outside, that structure is describable as one that repeatedly reduces future free energy at a maintenance cost lower than its benefit.

This also explains why abstraction emerges. Abstraction is not an additional faculty imposed on memory. It is a natural result of memory dynamics under finite resources. When several experiences repeatedly recruit overlapping circuits, those circuits become easier to reactivate and more energetically efficient. At the macro level, this appears as a structure that reduces free energy across situations.

Define the observer-level value of an abstract structure a as

$$V^{obs}(a) = \sum_j w_j \Delta F_a^{obs}(e_j) - C_{\mathrm{maintain}}(a). \tag{51}$$

If

$$V^{obs}(a) > 0, \tag{52}$$

then the abstract structure will look, from the modeler's perspective, worth preserving.

Therefore:

> An abstraction is a repeatedly recruited, locally strengthened structure that becomes cheaper to reactivate; in observer-level terms, it is a compressed structure that reduces free energy across multiple experiences by more than its maintenance cost.

7 The Brain Does Not Minimize Immediate Effort Alone

A common misunderstanding is that the brain simply chooses the easiest or least effortful path. This is only partly correct. A more precise claim is:

> Repeated stimulation tends to leave circuits that become easier and cheaper to reactivate. From the macro-level modeling perspective, these are the circuits with lower accumulated free energy over the long run. They are not selected by neurons through explicit optimization; they emerge from plasticity, threshold adaptation, and energetic constraints.

Many advanced abilities are costly at the beginning: learning mathematics, programming, language, or abstract modeling is initially demanding. Once a stable circuit is formed, however, future processing of similar problems becomes substantially cheaper.

The reason is not simply that the system has stored a symbolic answer. The learned circuit changes the mapping from future input to error:

$$\varepsilon_C(t|m_C) = o_t - g_C(m_C, W_C), \tag{53}$$

and therefore changes the future activation cost:

$$E_C(t|m_C) = E_{\mathrm{base}} + \chi\, \varepsilon_C(t|m_C)^\top \Pi_C \varepsilon_C(t|m_C). \tag{54}$$

As m_C and W_C become better priors for a class of situations, ε_C decreases. The long-term path becomes cheaper because the circuit no longer needs to re-create the same inference from scratch.

The brain therefore does not simply select

$$\text{the currently easiest path.} \tag{55}$$

It selects

$$\text{the path with lower accumulated long-term cost.} \tag{56}$$

Formally:

$$C^* = \arg\min_C \int_{t_0}^{t_1} [F_C(t) + \lambda_E E_C(t) + \lambda_M M_C(t) + \lambda_C C_C(t)]\, dt. \tag{57}$$

8 Overload, Fatigue, and Attentional Collapse

Let the current total cost rate be

$$L(t) = F_t + \beta \mathcal{G}_t + \lambda_E E_t + \lambda_A A_t + \lambda_M M_t + \lambda_C C_t. \tag{58}$$

Let the current resource boundary be

$$R_{\text{total}}(t). \tag{59}$$

When

$$L(t) < R_{\text{total}}(t), \tag{60}$$

the system can maintain stable inference, memory integration, and action control.

When

$$L(t) \geq R_{\text{total}}(t), \tag{61}$$

the system enters overload.

The brain may then adopt protective strategies: lowering input gain, narrowing attention, falling back to familiar circuits, reducing abstract reasoning, reducing higher-order integration, or producing fatigue, brain fog, dizziness, and avoidance.

An effective temperature can be introduced:

$$T_{\text{eff}}(t) = T_0 \max\left(1, \frac{L(t)}{R_{\text{total}}(t)}\right). \tag{62}$$

As load approaches or exceeds the resource boundary, noise increases and belief updating becomes less stable:

$$d\mu_t^k = -\Gamma_\mu \nabla_\mu F_t^k dt + \sqrt{2\Gamma_\mu T_{\text{eff}}(t)} dW_t^{\mu,k}. \tag{63}$$

This provides a formal model of cognitive overload.

However, caution is required. Dizziness, fatigue, and brain fog in real organisms can involve vestibular function, blood pressure, autonomic regulation, metabolic state, inflammation, sleep, and other physiological factors. The present model should not be treated as a unique causal account of these phenomena. It is a testable cognitive-level explanatory hypothesis.

9 The Higher-Order Observer as Meta-Control

In this framework, the higher-order observer should not be interpreted as a little person inside the brain watching everything.

More rigorously, it is a set of slow meta-control variables that regulate lower-level inference.

Define the meta-control state as

$$\theta_t = \{\lambda_E, \lambda_A, \lambda_M, \lambda_C, \Gamma_\mu, T_{\text{eff}}, \kappa\}. \tag{64}$$

These variables regulate resource cost weights, belief update speed, attentional precision, exploration noise, the competition temperature among candidate paths, and higher-order control gain.

The meta-control objective can be written as

$$J(\theta) = \mathbb{E}\left[\int_0^T \ell_t(\theta)dt\right]. \tag{65}$$

A meta-learning dynamics can be written as

$$d\theta_t = -\eta_\theta \nabla_\theta J(\theta_t)dt + \sigma_\theta dW_t^\theta. \tag{66}$$

Thus, the rigorous version of the higher-order observer is:

> A meta-control system that modulates precision, cost weights, update gains, and effective temperature.

Its role is not to mysteriously "see consciousness", but to redistribute finite resources among candidate cognitive paths so that the system tends to remain on lower long-term cost trajectories.

10 Unified Model

The preceding elements can be summarized as a resource-constrained active inference system:

$$\begin{aligned}
d\mu_t^k &= -\Gamma_\mu \nabla_\mu F_t^k dt + \sqrt{2\Gamma_\mu T_{\text{eff}}(t)} dB_t^{\mu,k} \\
dz_t^k &= \frac{-z_t^k - \ell_t^k}{\tau_z} dt + \sigma_z dB_t^{z,k} \\
\rho_t^k &= \frac{\exp(z_t^k / T_{\text{eff}})}{\sum_j \exp(z_t^j / T_{\text{eff}})} \\
dm_i &= [-\lambda_i m_i + \alpha_\varepsilon \|\Pi_t \varepsilon_i\| + \alpha_V V_i(t) + \alpha_N \text{Novelty}_i(t) + \alpha_c \, \text{coact}_i(t) - \alpha_E E_i(t)] \, dt + \sigma_m dB_t^{m,i} \\
d\theta_t &= -\eta_\theta \nabla_\theta J(\theta_t) dt + \sigma_\theta dB_t^\theta.
\end{aligned} \tag{67}$$

where

$$\hat{o}_t^k = G_k(m_t, \mathbf{W}_t^k), \tag{68}$$

$$\varepsilon_t^k = o_t - \hat{o}_t^k, \tag{69}$$

$$F_t^k = \frac{1}{2} (\varepsilon_t^k)^\top \Pi_t^k \varepsilon_t^k + \frac{1}{2} \lambda_W \|\mathbf{W}_t^k\|^2 + \text{Complexity}_t^k, \tag{70}$$

$$E_t^k = E_{\text{base}} + \chi (\varepsilon_t^k)^\top \Pi_t^k \varepsilon_t^k + E_{\text{plastic}}^k(t), \tag{71}$$

$$\ell_t^k = F_t^k + \lambda_E E_t^k + \lambda_A A_t^k + \lambda_M M_t^k + \lambda_C C_t^k - \lambda_V V_t^k - \lambda_{IG} IG_t^k. \tag{72}$$

Equivalently, when the logits are not modeled explicitly, path selection can be written directly as a softmax over negative cost:

$$\rho_t^k = \frac{\exp(-\ell_t^k / T_{\text{eff}})}{\sum_j \exp(-\ell_t^j / T_{\text{eff}})}. \tag{73}$$

The microscopic variables that make memory into a prior can be appended as

$$\dot{\mathbf{W}}_t^k = \eta_W q_t^k \Pi_t^k \varepsilon_t^k \phi(m_t)^\top - \lambda_W \mathbf{W}_t^k + \sigma_W \xi_W^k, \tag{74}$$

$$\tau_{\text{th}} \, dV_{\text{th},t}^k = \kappa_{\text{th}} \left(r_t^k - r_{\text{target}} \right) dt. \tag{75}$$

The model integrates the following closed chain:

$$\begin{aligned}
&\text{finite resources} \to \text{neural integration} \to \text{prediction error} \to \text{plasticity and adaptive thresholds} \\
&\to \text{priors} \to \text{lower future error and energy} \\
&\to \text{cognitive path cost} \to \text{multi-path competition} \to \text{memory dynamics} \\
&\to \text{abstraction} \to \text{meta-control} \to \text{adaptive intelligence}.
\end{aligned} \tag{76}$$

11 Potential Empirical Predictions and Operationalization

The model becomes scientifically useful only if its latent variables can be connected to observable quantities. The cost weights $\lambda_A, \lambda_C, \lambda_M$ and the effective temperature T_{eff} should not be treated as directly observed neural quantities. They are latent variables to be estimated by fitting behavioral and physiological data.

Several empirical predictions follow.

First, under dual-task load, time pressure, sensory noise, or sleep deprivation, the model predicts an increase in effective temperature. Candidate-path selection should therefore become more stochastic and higher entropy:

$$P(k) \propto \exp\left(-\frac{\mathcal{C}_k}{T_{\text{eff}}}\right). \tag{77}$$

As T_{eff} increases, choice distributions should flatten, reaction times should become more variable, and reliance on familiar low-cost strategies should increase.

Second, when control cost increases, the system should allocate less control to high-effort strategies even when those strategies are more accurate. This predicts larger task-switching costs, higher Stroop-like interference, reduced working-memory maintenance, and more habitual responding. Candidate empirical proxies include response time, error rate, subjective effort ratings, pupil diameter, frontal midline theta, and dACC/PFC BOLD activity.

Third, the local-repetition retention rule predicts that repeatedly co-activated traces should be more likely to remain reusable and to be reactivated later, especially when they improve subsequent prediction or action. This can be tested with delayed recall, recognition, sleep-dependent consolidation, representational reinstatement, and transfer tasks. The modeler may describe the successful trace as one that reduces future free energy, but the biological mechanism is local plasticity and consolidation, not explicit future-value computation.

Fourth, the abstraction rule predicts that shared structure across tasks should promote compressed representations. Under resource stress, abstraction depth should decrease: participants should rely more on concrete familiar exemplars and less on higher-level rules. In computational models, this should appear as a shift from higher-level latent variables toward lower-level cached policies.

These predictions do not prove that the biological brain exactly minimizes $\mathcal{C}_{path}$. They make the framework falsifiable: if manipulations of load, control cost, repetition, consolidation, or memory value do not change choice entropy, control allocation, retention, or abstraction in the predicted directions, the model must be revised.

12 What Can Be Proved and What Remains Hypothetical?

Some parts of the framework are standard or direct mathematical results:

Claim	Status
Variational free energy upper-bounds negative log evidence	Provable
Minimizing free energy approximates Bayesian posterior inference	Provable
Resource constraints yield a Lagrangian path functional	Provable
Softmax weights arise from accumulated cost	Provable
The continuous-time limit yields replicator dynamics	Provable
Logits guarantee legal probability weights	Provable
Gradient flow monotonically decreases free energy	Provable under specified conditions
Observer-level memory retention can be represented by a threshold rule	Derivable under net-benefit modeling
Abstract structure retention generalizes the observer-level memory rule	Derivable from that model

Other parts remain hypotheses:

Claim	Status
The biological brain exactly minimizes this cognitive path-cost functional	Hypothesis
The specific forms of energetic, attentional, mnemonic, and control costs are unique	Hypothesis
Higher-order meta-control maps cleanly onto a specific brain region	Requires empirical support
Effective temperature precisely represents cognitive overload	Effective model
Dizziness is directly caused by cognitive load exceeding a resource boundary	Not established
Biological memory computes future free-energy benefit exactly	Rejected as a literal mechanism
Abstraction is always equivalent to low-free-energy compression	Strong theoretical hypothesis

The framework should therefore not be read as a proof that the brain literally works this way. It is better understood as:

> A computable, simulable, and prediction-generating normative model of cognition.

13 Final Thesis

The whole theory can be condensed into one statement:

> The brain does not search for truth through unlimited computation. Under finite resources, it settles into low-action cognitive paths through local neural dynamics. Neural integration supplies the microscopic dynamics, repeated activation and plasticity supply the mechanism of memory, free-energy minimization supplies the observer-level description of error reduction, and the higher-order observer, understood as a meta-control system, reallocates resources among candidate paths. When cognitive load exceeds the system boundary, low-action paths become difficult to maintain, and the system may show fatigue, attentional collapse, failure of higher-order integration, and cognitive instability.

In a more academic form:

> The brain may be understood as a resource-constrained cognitive path optimization system. Under finite energetic, attentional, temporal, and mnemonic resources, local neural dynamics make repeatedly useful perceptual, mnemonic, and action trajectories easier to reactivate; at the macro level, these trajectories can be modeled as minimizing accumulated path cost and variational free energy while preserving adaptive self-environment coupling.

This is the core of the resource-constrained cognitive path optimization framework.

References

Friston, K. (2010). The free-energy principle: a unified brain theory? *Nature Reviews Neuroscience*, 11, 127–138.

Lieder, F., & Griffiths, T. L. (2020). Resource-rational analysis: Understanding human cognition as the optimal use of limited computational resources. *Behavioral and Brain Sciences*, 43, e1.

Parr, T., Pezzulo, G., & Friston, K. J. (2022). *Active Inference: The Free Energy Principle in Mind, Brain, and Behavior.* MIT Press.

Rissanen, J. (1978). Modeling by shortest data description. *Automatica*, 14(5), 465–471.

Schmidhuber, J. (2009). Driven by compression progress: A simple principle explains essential aspects of subjective beauty, novelty, surprise, interestingness, attention, curiosity, creativity, art, science, music, jokes. *Dagstuhl Seminar Proceedings.*

Shenhav, A., Botvinick, M. M., & Cohen, J. D. (2013). The expected value of control: An integrative theory of anterior cingulate cortex function. *Neuron*, 79(2), 217–240.